Proposal 7: The Cut-to-Box task

goal-directed release timing as a probe of internal pendulum dynamics

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Abstract

This note specifies one behavioral paradigm. A pendulum swings; on a button press the participant cuts the string, and the released bob follows a projectile path that should land in a target box. The only action is *when to cut*, a one-dimensional control. The task is diagnostic for a specific reason: the release state (θ, ω) is mapped to a landing point through a ballistic switch, and this map is *non-monotonic* in release phase, so "release later = land farther" is wrong. A participant who only tracks screen position cannot solve it; one who represents the full state and the tangential launch velocity can. Moving the box sweeps the value-maximizing release phase across the phase plane, which separates a *fixed dynamics geometry* (the pendulum's orbits and fixed points) from a *goal-dependent value geometry* (where releasing succeeds). This note fixes the stimulus, the conditions, the behavioral readouts, and the qualitative signatures that distinguish candidate strategies. The value model that formalizes those signatures is the companion Proposal 8; the significance and related work are Proposal 9.

1 Purpose

We want a task in which an action, not a prediction, reveals what dynamics a person is using. Passive prediction ("where will the bob be?") underdetermines the internal model, because many approximations agree on a short rollout. Acting toward a goal forces a commitment: the participant must convert whatever pendulum model they hold into a single motor decision whose payoff we can score.

Cut-to-Box is the cleanest such task we have found. The action is **one-dimensional** (the release time), so the entire decision is a point on the swing. The payoff is **analytic** (the post-release flight is a textbook projectile), so the optimal release is computable and the participant's error is interpretable. And the mapping from release phase to outcome is **non-monotonic and goal-dependent**, so simple heuristics make distinctive, falsifiable mistakes. The single question is:

> Does a person choose release timing by valuing the post-release ballistic outcome under an internal pendulum model, or by a position/timing heuristic that ignores the launch-velocity geometry?

2 The task

A pendulum of length L hangs from a pivot at height H and swings at a fixed amplitude. A box sits on the ground at horizontal distance D. The participant holds a *cut* key; releasing it cuts the string at that instant, and the bob leaves on a projectile path. The flight is occluded (the bob disappears at release and reappears at landing) so that the decision must come from an internal model rather than online visual tracking. Feedback is whether the bob landed in the box, plus its landing point.

A trial runs: observe the swing for 1–2 periods; the box appears (the goal cue); the participant cuts at a chosen moment; the flight is occluded; landing feedback is shown.

3 Why release timing is diagnostic

At the instant of release the bob has tangential speed $v = \omega L$ along the arc. With the pivot at $(0, H)$ and θ measured from the downward vertical, the release state (θ, ω) fixes both the launch position and the launch

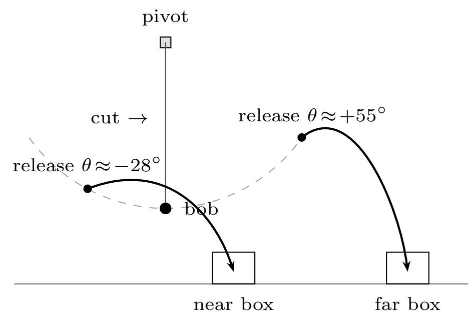

Figure 1: The Cut-to-Box task. The bob swings on the dashed arc; cutting the string at a chosen instant launches it on a projectile path (solid arrows) toward a box. The same pendulum reaches a near box and a far box only by cutting at *different* release phases. The flight is occluded in the experiment; it is drawn here for illustration.

velocity vector,

$$x_0 = L\sin\theta, \quad y_0 = H - L\cos\theta, \qquad v_x = \omega L\cos\theta, \quad v_y = \omega L\sin\theta. \tag{1}$$

The bob then falls as a projectile; taking the descending root to the box height y_b,

$$t_{\text{land}} = \frac{v_y + \sqrt{v_y^2 + 2g(y_0 - y_b)}}{g}, \qquad x_{\text{land}} = x_0 + v_x\, t_{\text{land}}. \tag{2}$$

Two features of this map make the task discriminating (Figure 2).

The landing map is non-monotonic in release phase. On a fixed swing (here $L = 1\,\text{m}$, $H = 2.2\,\text{m}$, amplitude 85°) the reachable landings run from -1.0 to $3.07\,\text{m}$, and the *maximum range is not at the bottom of the swing* but near $\theta \approx 31°$: releasing on the upswing gives $v_y > 0$, which lengthens the flight. So "cut later to throw farther" fails over part of the range, and a mid-range box is reachable by *two* different release phases whose value ridges differ in width (one is more tolerant of timing error).

Two geometries live on the same phase plane. The pendulum contributes a *dynamics geometry* that does not depend on the goal: the center fixed point at $(0, 0)$, the saddles at $(\pm\pi, 0)$, the closed energy orbits, and the separatrix (gray in Figure 2b,c). The box contributes a *value geometry*: the set of release states that land in it, i.e. the bright ridge of $V_g(\theta, \omega)$, which *moves* as the box moves. Solving the task is finding where the current orbit intersects the value ridge. As D grows from 0.8 to 2.4 m the optimal release phase shifts from $\theta^\star \approx -28°$ to $\theta^\star \approx +55°$. The same physical state thus acquires different value under different goals — which is exactly the quantity the participant must represent.

4 Conditions and manipulations

- **Box distance** D (about six levels, near → far, including the two-solution mid-range). Predicts a specific, non-monotonic shift of release phase (Figure 2a).
- **Orbit energy / amplitude** (small vs. large swing). Changes the reachable range; on a small swing the far box is *unreachable*, which sets up the reachability test below.
- **Occlusion of the flight** (visible vs. fully occluded). Forces reliance on the internal model; the occluded condition is the primary one.
- **Recalibration block (optional)** changing L or g. Rescales the whole map and tests whether participants transfer a model or relearn a stimulus–response table.

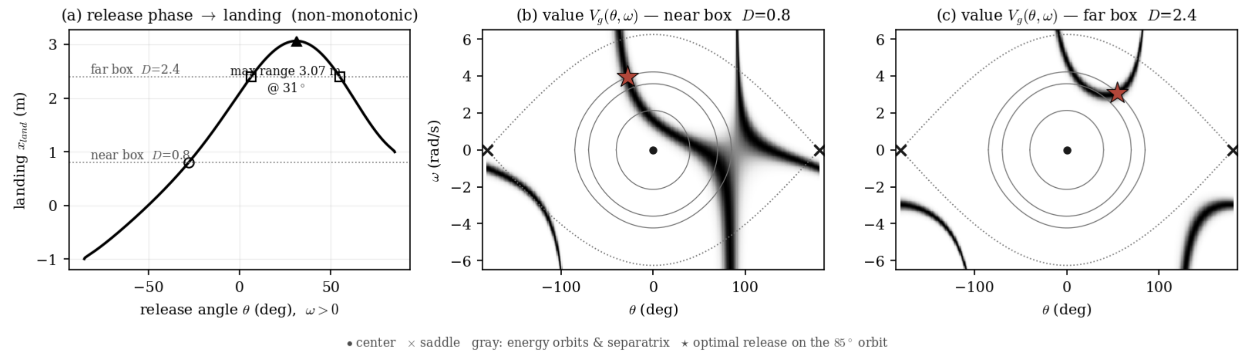

Figure 2: Value structure (numerical; $L = 1\,\mathrm{m}$, $H = 2.2\,\mathrm{m}$, amplitude 85°). **(a)** Landing distance vs. release phase is non-monotonic: maximum range 3.07 m at $\theta \approx 31°$, not at the bottom; near and far boxes (open circle, open square) need different release phases, and mid-range boxes have two solutions. **(b,c)** Value $V_g(\theta, \omega)$ over the phase plane. The dynamics geometry (gray orbits and separatrix; • center, × saddle) is identical in both panels; the value ridge (dark) moves with the box, and the optimal release on the 85° orbit ($\star$) shifts accordingly.

Reachability and a natural extension. When the box lies beyond the current orbit's maximum range, the correct behavior is not to cut at all but first to *pump* the swing to a larger orbit and then release. This "pump-then-release" variant chains orbit dynamics, energy injection, the release switch, and the goal in one task; we flag it here and develop it in the model note (Proposal 8).

5 Behavioral readouts and predicted signatures

The primary readouts are the **release-phase distribution** per condition, the **signed landing error** as a function of D (over- vs. undershoot), the success rate, the decision time, the trial-to-trial variability at repeated states, and the rate of *declared-unreachable* responses on small orbits. Three strategies leave distinct fingerprints; the formal models are in Proposal 8.

- **Full internal model.** Release phase tracks the analytic optimum across D, *including* the non-monotonic regime, and on two-solution boxes prefers the wider (more timing-robust) value ridge.
- **"Later = farther" heuristic.** Release phase increases monotonically with D; this overshoots systematically once D requires release past the $\theta \approx 31°$ range peak. This is the strongest separation from the full model.
- **Fixed small-angle model.** Underestimates tangential speed and phase at large amplitude, biasing landings short, with error that grows with swing amplitude.

Lowering evidence (short observation, low contrast) should broaden the release distribution and pull it toward an amplitude-averaged guess; the occlusion manipulation should hurt the heuristic least and the full model most informatively (because only the model needs the hidden launch geometry).

6 Pilot and validation plan

A pilot first samples D and release phase on a coarse grid to locate where the candidate strategies diverge most — the far-box, large-amplitude corner where the non-monotonicity bites — and then densifies sampling there. A workable first design crosses ∼6 distances × 2 amplitudes × 2 occlusion levels with ∼20 repetitions (≈480 trials/participant), $n \approx$ 20–30. Sanity checks before running: verify the rendered projectile obeys Eq. (2); confirm that the analytic optimal-release curve (Figure 2a) is reproducible from the stimulus engine; and confirm that at least one box distance is reachable only by releasing past the range peak, so the heuristic and the model must disagree.

7 Summary

Cut-to-Box reduces goal-directed physical action to a single decision — when to cut — whose optimal value is analytic and whose landing map is non-monotonic in release phase. That non-monotonicity, plus the goal-driven motion of the value ridge across a fixed dynamics geometry (Figure 2), is the lever: a position/timing heuristic and a full state-plus-launch-velocity model make different, measurable release choices. The companion Proposal 8 turns these signatures into a fitted value model; Proposal 9 places the task in the literature.